RYAN P. SEAMAN

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SUMMARY

Computational biologist and data scientist pursuing an MMSc in Biomedical Informatics at Harvard Medical School. Experienced in bioinformatics pipelines, cloud infrastructure, machine learning, and single-cell genomics, with current thesis work applying multi-agent LLM systems to automated data visualization.

EDUCATION

Harvard Medical School Boston,MA
MMSc. Biomedical Informatics *Expected Graduation May 2027*

Colby College Waterville, ME
B.A. Computational Biology January 2023
Varsity Soccer, 4 years

EXPERIENCE

Graduate Research Assistant, HIDIVE Lab December 2025 – Present
Harvard Medical School, Dr. Nils Gehlenborg Boston, MA

- Designing and developing VitessceGen, an AI-assisted tool that leverages a multi-agent LLM architecture to automatically generate validated JSON configurations for the Vitessce visual integration tool
- Engineering a system of specialized agents (retrieval, code generation, validation) that collaborate to transform heterogeneous biomedical datasets into interactive, multi-panel visualizations
- Evaluating system performance against expert-authored configurations across spatial transcriptomics, single-cell RNA-seq, and imaging datasets from the Human BioMolecular Atlas Program (HuBMAP)

Bioinformatician, Comparative Genomic and Data Science Core January 2023 – August 2025
MDI Biological Laboratory Bar Harbor, ME

- Developed and implemented data analysis tools that empower researchers to explore and analyze their own data, improving research efficiency
- Built and deployed an open-source single-cell RNA sequencing pipeline for downstream analysis, ensuring rigorous, reproducible data analysis
- Architected and maintained cloud infrastructure (GCP/AWS) for secure analysis and data storage
- Led educational programs, including internal trainings and courses for the broader research community on bioinformatics tools and techniques

SKILLS and TECHNIQUES

Genomics: RNAseq, scRNAseq, 10x Visium, 10x Xenium, 10x Multiome (scRNAseq + scATACseq)
Coding/Compute: Python, R, Bash, Nextflow, Groovy, Git, Google Cloud Platform, Amazon Web Services, Slurm
Other skills: teaching/mentoring, computing/research ethics

PUBLICATIONS

Seaman RP, et al. (2024). A cloud-based training module for efficient de novo transcriptome assembly using Nextflow and Google Cloud. *Brief Bioinform*.

García-García, D, Knapp, D, Kim, M, Jamwal, K, Fuqua, H, **Seaman, RP**, et al. (2025). The essential role of connective-tissue cells during axolotl limb regeneration. *bioRxiv.*

Paulmann A, Cox MD, Boewer T, Somers HM, Fuqua H, **Seaman RP**, et al. (2025). Sodium-glucose co-transporter 2 inhibition improves age-dependent kidney microvascular rarefaction. *Kidney Int*.

Sandhi S, Somers H, Cox M, Nobrega C, **Seaman RP**, et al. (2026). Sexually dimorphic response to dietary restriction-induced longevity and muscle rejuvenation in *Nothobranchius furzeri*. *bioRxiv*.